

100 Days of Azure DevOps

Day 35 Handout — Two complete worlds, one pointer. Rollback is a light switch — only if you kept the previous artifact

4 - Continuous Delivery · Personal lab only · Educational content · Not a sales pitch

Architecture A — blue-green mapped onto App Service slots

Idea	App Service lab map	Not this
Blue (live)	Production slot + production hostname	A single in-place zip overwrite
Green (candidate)	staging slot, smoked, still idle to users	A branch named green with no second host
Cutover	Swap (VIP flip) — 100% in one move	A 5% traffic ramp (that is canary, Day 36)
Rollback	Swap back; same two worlds	Rebuild last week's commit from a laptop
AKS later	Two Deployments + Service/Ingress flip	Not today's cluster bill

Architecture B — when blue-green is the right switch

Constraint	Prefer blue-green	Prefer something else
Need instant rollback	Yes — flip the pointer	Rolling (Day 37) rolls back in batches
Cannot pay for two worlds	Slots on one plan (lab-sized)	In-place deploy + accept downtime
Need a small slice of users first	No — that is canary	Day 36 traffic % or a feature flag
N and N-1 cannot coexist	Green is isolated; cutover is atomic	Do not roll; do not mix versions on one fleet

Architecture C — pitfalls that turn rollback into archaeology

Pitfall	What you see	Fix
No previous drop	Swap back lands old slot with new bits already	Keep artifact drop N-1 for a written window (lab: 7 days)
Vanity health	200 on / that never touches a dependency	Smoke as a user: a path that hits data if you have data
Called it blue-green	One slot, one deploy, a blog title	If there is no idle world, there is no switch
Schema change first	Green needs a column blue cannot read	Expand/contract the data before the flip

One-liner to remember

Blue-green means two worlds; only one takes traffic — rollback is a light switch, not an archaeology dig.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure / Azure DevOps only. If slots are unavailable, the rollback document still ships. No employer diagrams.

- 1 Draw blue = production slot, green = staging, pointer = swap. One box each in docs/rollback.md (or a comment sketch).
- 2 Write docs/rollback.md with three numbered steps: (1) swap back staging/production, (2) verify a real URL, (3) keep previous pipeline artifact 7 days.
- 3 If S1+ slots exist from Day 34: deploy N to green, smoke, swap (blue↔green), then swap back using only the doc.
- 4 Record the artifact run ID you would keep as N-1. CD must download that drop — not rebuild.
- 5 Add a sentence on schema: if green needs a breaking column, the switch is unsafe until expand/contract.
- 6 Do not stand up AKS for this metaphor. Slots are the lab-sized two worlds.

rollback.md — the switch, not a scavenger hunt

```
# docs/rollback.md
# Blue = production slot. Green = staging. Pointer = swap.
# 1. az webapp deployment slot swap -g <rg> -n <app> --slot staging --target-slot production
#    (swap back is the same command after another swap, or swap again)
# 2. curl -fI https://<app>.azurewebsites.net/ (use a path that hits a dependency if you have one)
# 3. Keep Pipeline artifact 'drop' from run <BuildId> for 7 days – do not rebuild that commit.
```

Done checklist

- Can distinguish blue-green (100% flip) from canary (ramp)
- docs/rollback.md has swap, verify, keep artifact
- Did not call a single-slot deploy blue-green
- No AKS spend today

Tomorrow — Day 36

Canary releases — a small slice of real traffic, a watch, and an abort that is faster than a debate.